Denote V be a Vector space over a field F.

Def. Let $S \subseteq V$ be a subset.

- Define linear-combinations of S as;

$$\{a_1 v_1 + \cdots + a_n v_n \mid n \in \mathbb{N}, a_i \in F, v_i \in S\}.$$

- A subset $S \subseteq V$ is said to be linearly independent iff for any finite subset $\{v_1, \cdots, v_n\} \subseteq S$,

$$\sum_{i=1}^{n} a_i v_i = 0 \Rightarrow a_1 = \cdots = a_n = 0.$$

Lemma. Let $S \subseteq V$ be a subset. Then,

$$\langle S \rangle = \{a_1 v_1 + \cdots + a_n v_n \mid n \in \mathbb{N}, a_i \in F, v_i \in S\}.$$

Prof. Denote the right term as $\overline{S}$. Clearly, $S \subseteq \overline{S} \leq V$.

Since $\langle S \rangle = \bigcap_{S \subseteq W \leq V} W$ is the smallest subspace containing S, $\langle S \rangle \leq \overline{S}$.

Conversely, $\overline{S} \leq \langle S \rangle$ because $\langle S \rangle$ contains S and closed under the operations.

Lemma. Let $S \subseteq V$ be a subset. TFAE;

a) S is linearly independent

b) If $v \in V$ is linear combination of S, then this representation is unique.

Prof. a) $\Rightarrow$ b) If there exist two representation of v by linear combination of S:

$$v = a_1 v_1 + \cdots + a_n v_n = b_1 v_1 + \cdots + b_n v_n$$

This implies that

$$(a_1 - b_1) v_1 + \cdots + (a_n - b_n) v_n = 0$$

That is, for all $1 \leq i \leq n$, $a_i - b_i = 0$.

b) $\Rightarrow$ a) Let finite vectors $v_1, \cdots, v_n \in S$ be given.

If $a_1 v_1 + \cdots + a_n v_n = 0$ for some $a_i \in F$,

then $a_1, \cdots, a_n$ must be zero, by uniqueness of representation of zero.

Thm. Let $\beta \subseteq V$ be a subset. TFAE;

a) β is basis of V

b) For any $x \in V$, there exists a unique representation of x by linear combination of β.

⌈Thm. If a finite subset $S \subseteq V$ generates V, then there exists a basis β of V in S.

Proof. If $S \subseteq \{0\}$, then $V = \langle S \rangle = \{0\}$. Thus, $\emptyset \subseteq S$ is a basis of V.

Suppose S contains a non-zero vector $u_1 \in S$. Then, for any $u_2 \in S \setminus \langle u_1 \rangle$,

$$a_1 u_1 + a_2 u_2 = 0 \Rightarrow a_1 = a_2 = 0$$

Because either $a_1 \neq 0$ or $a_2 \neq 0$, the cancelation gives $u_2 \in \langle u_1 \rangle$, contradiction.

Thus $\{u_1, u_2\} \subseteq S$ is linearly independent. (If $S \setminus \langle u_1 \rangle = \emptyset$, then $\{u_1\}$ is basis of V).

Similarly, for $u_3 \in S \setminus \langle \{u_1, u_2\} \rangle$, $\{u_1, u_2, u_3\} \subseteq S$ is linearly independent.

Since S is finite, this process will terminate in finite iteration.

That is, for some $n \in \mathbb{N}$, $S \subseteq \langle \{u_1, \cdots, u_n\} \rangle$. This $\{u_1, \cdots, u_n\}$ is basis of V, by

$$V = \langle S \rangle \subseteq \langle \{u_1, \cdots, u_n\} \rangle.$$

⌈Replacement Theorem.

Suppose that $G \subseteq V$ generates V, $L \subseteq V$ is linearly independent, with $\#(G) = n$, $\#(L) = m$.

Then, $m \leq n$ and there exists $H \subseteq G$ s.t. $\#(H) = n - m$ and $\langle L \cup H \rangle = V$.

Proof. Using Induction for m.

If $m = 0$, (that is, $L = \emptyset$), then take $H = G$.

Now suppose for $m \in \mathbb{N}$, the statement holds.

Let $L = \{v_1, \cdots, v_m, v_{m+1}\}$ be a linearly independent subset of V,

and G be a generating subset of V with $\#(G) = n$.

Since subset of linearly independent set is linearly independent, $\{v_1, \cdots, v_m\} \subseteq L$ is linearly indep.

Now, there exists a subset $\{u_1, \cdots, u_{n-m}\} \subseteq G$ such that

$$\langle \underbrace{\{v_1, \cdots, v_m\} \cup \{u_1, \cdots, u_{n-m}\}}_{=K} \rangle = V$$

That is, there exists a linear combination of K to v_{m+1} as

$$\sum_{i=1}^{m} a_i v_i + \sum_{i=1}^{n-m} b_i u_i = v_{m+1}$$

If $n - m = 0$, then $v_{m+1} = \sum_{i=1}^{m} a_i v_i$, it is contradiction to linearly independent.

Now, $n \geq m+1$. And b_i are not all zero; if all zero, then it is contradict to linearly independent.

For some $1 \leq j \leq n-m$, take $b_j \neq 0$. Then,

$$u_j = b_j^{-1} \cdot \left(v_{m+1} - \sum_{i \neq j} b_i u_i - \sum_{i=1}^{m} a_i v_i\right) \in \langle L \cup H \rangle$$

where $H = \{u_1, \cdots u_{j-1}, u_{j+1}, \cdots, u_{n-m}\}$. Since $K \subseteq L \cup H$ and $\langle K \rangle = V$, $\langle L \cup H \rangle = V$.

⌈Corollary, If V has a finite basis, then every basis of V has same cardinality.